Ordinary Differential Equations

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1 Introduction

An ordinary differential equation (ODE) is an equation involving an unknown function and its ordinary derivatives with respect to **a single independent variable**. The goal of solving an ODE is to find a function that satisfies the given equation.

For example, consider

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \cos(t),$$

we know

$$x(t) = \sin(t) + C,$$

where C is an arbitrary constant. This example illustrates an important point that knowing the derivative of a function does not necessarily determine the function **uniquely**. Now, suppose we given the initial condition

$$x(0) = 2.$$

Substituting $t = 0$ into the general solution gives

$$x(0) = \sin(0) + C = C.$$

Since $x(0) = 2$, we obtain $C = 2$. Therefore, the unique solution satisfying both the ODE and the initial condition is

$$x(t) = \sin(t) + 2.$$

In general, solving an ODE can be much more complicated than finding an antiderivative. Nevertheless, integration provides a fundamental connection between a differential equation and its solution.

More generally, we can specify a condition of the form

$$x(t_0) = x_0,$$

where t_0 is a given initial point and x_0 is the corresponding value of the unknown function. The is called an **initial condition**. This leads naturally to the concept of an **initial value problem (IVP)**, in which a differential equation is accompanied by a prescribed value of the unknown function at an initial point.

2 Initial Value Problem

The initial value problem for a first-order ordinary differential equation

$$\begin{cases} y' = f(t, y(t)), & t \in [a, b] \\ y(a) = y_a \end{cases}$$

For example, **the logistic equation** $y' = ay(1 - y)$.
In real model, y usually represents the population.

$$\text{I.V.P} \quad \begin{cases} y' = ay(1 - y), & t \in [0, T] \\ y(0) = y_0 \end{cases} \qquad \longrightarrow \qquad y(t) = 1 - \frac{1}{1 + \frac{y_0}{1-y_0} e^{at}}, \text{ if } y_0 \neq 1.$$

A wide majority of interesting D.E. have no closed-form solution. A much more common scenario is a D.E. with no explicit solution formula. For example,

$$\begin{cases} y' = 3e^{\frac{t^2}{2}} - t^2 - 2 \\ y(0) = 1 \end{cases}$$

When an ordinary differential equation (ODE) has no closed-form solution, we cannot express its solution using standard analytical formulas. In such cases, we must turn to **numerical methods** to approximate the solution at discrete points.

3 Euler's Method

Euler's method uses the derivative at the current point to estimate the solution at the next point. Recall the definition of a derivative

$$y'(t) = \lim_{h \to 0} \frac{y(t + h) - y(t)}{h}.$$

For a sufficiently small step size h, we approximate the derivative by a forward difference

$$y'(t) \approx \frac{y(t + h) - y(t)}{h}.$$

Rearranging gives

$$y(t + h) \approx y(t) + hy'(t).$$

Since $y'(t) = f(t, y(t))$, we obtain Euler's formula

$$y_{n+1} = y_n + hf(t_n, y_n),$$

3.1 Lipschitz constant

Definition: A function $f(t, y)$ satisfies the Lipschitz condition in the variable y on the rectangles $S = [a, b] \times [\alpha, \beta]$ if there exists a constant L (called the **Lipschitz constant**) such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2| \quad \text{for all } (t, y_1), (t, y_2) \text{ in } S.$$

Note: A useful condition is that if the partial derivative $\frac{\partial f}{\partial y}$ exists and is bounded on a suitable region, then f is Lipschitz continuous in y there. According to Mean value theorem

$$\frac{f(t, y_1) - f(t, y_2)}{y_1 - y_2} = \frac{\partial f}{\partial y}(t, c)$$

where c is between y_1 and y_2. Then L can be taken as the maximum of $|\frac{\partial f}{\partial y}(t, c)|$ on S.

Example 1: A function that satisfies the condition. Consider

$$f(t, y) = 2y.$$

Then

$$|f(t, y_1) - f(t, y_2)| = 2|y_1 - y_2|.$$

Therefore, f is **globally** Lipschitz continuous in y, with $L = 2$.

Example 2: A function that is **locally** Lipschitz. Consider

$$f(t, y) = y^2.$$

We have

$$|y_1^2 - y_2^2| = |y_1 - y_2||y_1 + y_2|.$$

If $|y_1|$, $|y_2| \leq M$, then

$$|y_1^2 - y_2^2| \leq 2M|y_1 - y_2|.$$

Thus, f is Lipschitz continuous on every bounded interval in y, and **locally** Lipschitz on the entire real line.

Example 3: A continuous function that is not locally Lipschitz. Consider

$$f(t, y) = \sqrt{|y|}.$$

At $y = 0$, compare $y_1 = 0$ and $y_2 = y > 0$. The Lipschitz condition would require

$$\sqrt{y} \leq Ly,$$

or equivalently,

$$\frac{1}{\sqrt{y}} \leq L.$$

As y approaches zero, the left-hand side becomes unbounded. No finite L can satisfy the condition in a neighborhood of zero. So f is continuous but not locally Lipschitz at $y = 0$.

3.2 Existence and Uniqueness Theorem (E.U.T)

Definition: Assume that $f(t,y)$ is Lipschitz in the variable y on $[a,b]\times[\alpha,\beta]$ and $y_a \in (\alpha,\beta)$. Then $\exists c \in (a,b)$ such that I.V.P. $\begin{cases} y' = f(t,y) \\ y(a) = y_a \end{cases}$ has exactly one solution $y(t)$ for $t \in [a,c]$. Moreover, if f is Lipschitz on $[a,b]\times(-\infty,\infty)$, then there exists a unique solution on $[a,b]$.

Example 1: Consider $\begin{cases} y' = y^2 \\ y(0) = 1 \end{cases}$.

Since $f(t,y) = y^2$ we know $\frac{\partial f}{\partial y} = 2y$. $L = \max|2y| = 20$ on $[0,2]\times[-10,10]$. E.U.T. guarantees $y(t)$ uniquely exists on $[0,c]$, $c>0$ but not guarantees on the entire $[0,2]$.

In fact, $y(t) = \frac{1}{1-t}$. So, the solution exists on $[0,c]$ for any $0<c<1$.

Note: Since$\frac{dy}{y^2} = dt$, we obtain $-\frac{1}{y} = t + C$. Substituting the initial condition, so $C = -1$. Thus, $-\frac{1}{y} = t-1$ which can be rewritten $y = \frac{1}{1-t}$.

Example 2: Consider $\begin{cases} y' = y^2 \\ y(0) = y_0 \end{cases}$.

From previous example we know that $f(t,y) = y^2$ is Lipschitz continuous on $[-a,a]$ for all $a>0$.

Case 1: $y_0 \equiv 0$. Then $y(t) = 0$ is a solution.

Case 2: $y_0 \neq 0$. Then $y(t) = \frac{1}{y_0^{-1} - t}$.

$$\begin{cases} \text{If } y_0 > 0, \text{ then } y(t) = \frac{1}{y_0^{-1}-t} \text{ for } t \in (-\infty, \frac{1}{y_0}) \\ \text{If } y_0 < 0, \text{ then } y(t) = \frac{1}{y_0^{-1}-t} \text{ for } t \in (\frac{1}{y_0}, \infty) \end{cases}$$

Example 3: Consider $\begin{cases} y' = \begin{cases} \sqrt{y}, y \geq 0 \\ 0, y \leq 0 \end{cases} \\ y(0) = 0 \end{cases}$.

Case 1: $y(t) = \begin{cases} \frac{(t-c)^2}{4}, t \geq c \\ 0, t \leq c \end{cases}$ is a solution for any $c \geq 0$.

Case 2: $y(t) \equiv 0$ is also a solution.

Thus, in this example, there are infinitely many solutions.

Note: $\frac{dy}{\sqrt{y}} = dt$. We can write $\int y^{-1/2} dy = \int 1 dt$. So, $2\sqrt{y} = t + C$. We have $y = \frac{(t+C)^2}{4}$. Set $C = -c$, we have $y = \frac{(t-c)^2}{4}$

3.3 Continuous dependence of solutions on initial conditions

Definition: Assume that $f(t,y)$ is Lipschitz in variable y on $S=[a,b]\times[\alpha,\beta]$. If $y(t)$ and $z(t)$ are solutions of $\begin{cases} y'=f(t,y) \\ y(a)=y_a \end{cases}$ and $\begin{cases} z'=f(t,z) \\ z(a)=z_a \end{cases}$, respectively, then

$$|y(t)-z(t)| \le e^{L(t-a)}|y(a)-z(a)|.$$

Proof: By the uniqueness of solution, $y(a)=z(a)$ implies $y(t)=z(t)$. We may assume $y(a)\neq z(a)$. In this case, the uniqueness implies $y(t)\neq z(t), \forall t$.
Define $u(t)=y(t)-z(t)$, then $u(t)>0$ or $u(t)<0$. Because $|y(t)-z(t)|$ only depends on $|u|$, we may assume $u>0$.

$$u'(t)=y'(t)-z'(t)=f(t,y(t))-f(t,z(t))$$

The Lipschitz condition implies

$$u' \le |u'| = |f(t,y(t)-f(t,z(t))| \le L|y(t)-z(t)| = L|u(t)| = Lu(t),$$

which means

$$(\ln u)' = \frac{u'}{u} \le L.$$

By the mean value theorem, $\frac{\ln u(t)-\ln u(a)}{t-a} \le L$ which simplifies to $\ln\frac{u(t)}{u(a)} \le L(t-a)$. Thus, $u(t)\le u(a)e^{L(t-a)}$.

3.4 Global error

Assume $f(t,y)$ has a Lipschitz constant L for the variable y and y_i is the solution of IVP $\begin{cases} y'=f(t,y) \\ y(a)=y_a \end{cases}$ $a \le t \le b$ at t_i, w_i is the one-step ODE solver withe local truncation error $e_i \le ch^{k+1}$. Then for $a\le t_i\le b$, the solver has global truncation error

$$g_i = |w_i - y_i| \le \frac{c}{L}h^k(e^{L(t_i-a)}-1).$$

Proof: Set $g_0=|w_0-y_0|=|y_a-y_a|=0$ and $g_1=e_1=|w_1-y_1|$.
Let $z(t)$ be the solution of $\begin{cases} y'=f(t,y) \\ y(t_1)=w_1 \end{cases}$, $t\in[t_1,t_2]$ and then $e_2=|w_2-z(t_2)|$.
By initial value continuous theorem, we have $|z(t_2)-y_2|\le e^{Lh}|w_1-y_1|$. Thus

$$\begin{aligned} g_2 &= |w_2-y_2| \\ &= |w_2-z(t_2)+z(t_2)-y_2| \\ &\le |w_2-z(t_2)|+|z(t_2)-y_2| \\ &\le e_2+e^{Lh}g_1 \\ &= e_2+e^{Lh}e_1. \end{aligned}$$

Similarly,

$$\begin{aligned} g_3 &= |w_3 - y_3| \le e_3 + e^{Lh} g_2 \le e_3 + e^{Lh} e_2 + e^{2Lh} e_1. \\ &\vdots \\ g_i &= |w_i - y_i| \le e_i + e^{Lh} e_{i-1} + e^{2Lh} e_{i-2} + \cdots + e^{(n-1)Lh} e_1. \end{aligned}$$

Assume the local truncation error $e_i \le c \cdot h^{k+1}$. Then

$$\begin{aligned} g_i &\le ch^{k+1}[1 + e^{Lh} + \cdots + e^{(i+1)Lh}] \\ &= ch^{k+1} \frac{e^{iLh-1}}{e^{Lh} - 1} \\ &\le ch^{k+1} \frac{e^{L(t_i - a)-1}}{Lh} \\ &= \frac{c}{L} h^k (e^{L(t_i-a)-1}). \end{aligned}$$

Note:
$i \cdot h = (t_i - a)$
$1 + x \le e^x$ which means $1 + (Lh) \le e^{Lh}$. Then $Lh \le e^{Lh} - 1$.

3.5 Euler's Method Error

Assume $y \in \mathbb{C}^2$ the exact solution at $t_{i+1} = t_i + h$ is

$$y(t_i + h) = y(t_i) + h \cdot y'(t_i) + \frac{h^2}{2} y''(c), t_i \le c \le t_{i+1}.$$

Assume $y(t_i) = w_i$ and $y'(t_i) = f(t_i, w_i)$. Then

$$y(t_{i+1}) = w_i + h \cdot f(t_i, w_i) + \frac{h^2}{2} y''(c).$$

Meanwhile, Euler's method says $w_{i+1} = w_i + h \cdot f(t_i, w_i)$ which means

$$e_{i+1} = |w_{i+1} - y(t_{i+1})| = \frac{h^2}{2} |y''(c)| \le \frac{M}{2} h^2, \text{ where } M \equiv \max_{[t_i, t_{i+1}]} y''(x).$$

Corollary: Assume $f(t, y)$ has a Lipshitz constant L in variably y. The global truncation error from Euler's method satisfies

$$|w_i - y_i| \le \frac{Mh}{2L} (e^{L(t_i - a)-1}), M = \max_{t \in [a,b]} y''(t).$$